

KENAR NARAYAKA

+44 7918 448877 kenar.narayaka@cs.ox.ac.uk linkedin.com/in/narayaka github.com/kxnar

Education

University of Oxford

MCompSciPhil Computer Science & Philosophy

Oct. 2025 – Jun. 2029

Christ Church

Loughborough College

*A-Level Mathematics, Further Mathematics, Computer Science (A*AA)*

Sep. 2023 – Jun. 2025

Loughborough, UK

Woodbrook Vale Secondary School

10 GCSEs (9 Grade 9s, 1 Grade 8); FSMQ Additional Mathematics (A)

Sep. 2018 – Jun. 2023

Loughborough, UK

Relevant Coursework

- Design & Analysis of Algorithms
- Discrete Mathematics
- Intro. Proof Systems
- Functional Programming
- Imperative Programming
- Probability
- Logic

Experience

Maths & Computer Science Tutor

April 2025 – Present

Independent

Remote / Loughborough, UK

- Delivered sustained 1-to-1 tutoring in mathematics and computer science for competitive university entrance exams, typically involving daily sessions and independent practice support
- Designed and delivered targeted preparation integrating calculus, algebra, probability, and Python-based problem solving for selective, exam-based admissions
- Developed structured revision plans, custom problem sets, and timed practice to improve reasoning speed, accuracy, and exam performance
- Supported successful admission to a competitive private university through focused exam preparation and consistent performance tracking
- Taught Python as a tool for reinforcing mathematical concepts, algorithmic thinking, and problem decomposition beyond the standard curriculum

Hitachi Astemo

June 2024 – August 2024

Backend Engineer Intern

Jakarta, Indonesia

- Developed internal workflow systems in PHP; fully replacing paper-based processes, improving traceability, reducing manual handling, and streamlining the HR system's workflows
- Designed and implemented a digital request pipeline using PHP and Fat-Free Framework, including form validation, PDF generation, automatic approval logic, and network printing
- Built REST APIs and administrative tools for high-volume records with strict consistency requirements, role-based access control, and reliable data handling
- Optimised SQL queries and ORM mappings to reduce latency, improve query efficiency, and support faster updates across large employee datasets

Projects

3D Fractal Simulator | *JavaScript, three.js, GLSL* | *Loughborough College*

Feb. 2025

- Developed a real-time 3D fractal visualiser for A-Level Computer Science Non-Examined Assessment using JavaScript, three.js, and GLSL fragment shaders with raymarching and cone-marching (vertex shader) techniques.
- Achieved 69/70 score on OCR CS A-Level NEA, highest in school history.
- Implemented configurable fractals including Mandelbulb, Mandelcube, and Juliabulb with adjustable parameters for scale, iteration depth, morph speed, and camera behaviour
- Optimised GPU performance through shader-level mathematical simplifications, reduced computational overhead, and dynamic fractal iteration scaling for smoother real-time rendering when factoring camera movement.
- Designed interactive UI for live parameter manipulation, making complex mathematical structures more accessible and easier to explore in real time.

Activities

CapitOx Quant Academy

Jan. 2026 – Present

University of Oxford

- Selected for a competitive programme run with the Oxford Man Institute focused on systematic investing
- Completed training in portfolio optimisation, factor models, and multi-signal strategies
- Developed and backtested example investment strategy using Python and C++